

SAN, Mali

WMO: 612770

Lat: 13.333N

Lon: 4.833W

Elev: 284

StdP: 97.96

Time Zone: 0.00 (E00)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	17.0	-4.1	2.8	30.5	-2.4	3.2	29.5	7.5	23.9	6.3	25.6	1.2	N/A	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.3	41.9	22.5	41.1	22.4	40.3	22.3	27.6	34.4	27.2	33.7	26.9	33.2	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.3	30.5	25.7	21.7	30.0	25.4	21.2	29.7	89.8	34.8	87.8	33.8	86.2	32.9	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.8	5.0	4.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	24.6	28.4	31.9	34.1	34.3	31.9	29.0	27.8	28.6	30.6	28.8	26.1
	DBStd	3.46	2.20	2.76	2.21	1.69	1.93	2.35	1.90	1.46	1.62	1.31	1.64	1.97
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7178	453	515	679	723	752	656	589	553	558	637	564	498
	CDD18.3	4137	195	282	421	473	494	406	331	295	308	379	314	239
	CDH23.3	56586	2105	3622	6430	7809	8048	6016	4086	3207	3568	5093	3981	2621
	CDH26.7	33649	1028	2135	4198	5437	5593	3737	2002	1308	1651	2868	2311	1381
(14) Wind	WSAvg	1.4	1.7	1.9	1.5	1.5	1.7	1.7	1.3	1.0	0.8	0.9	1.2	1.6
(15) Precipitation	PrecAvg	693	1	0	2	8	36	92	178	227	132	31	1	0
	PrecMax	895	25	5	32	54	96	179	327	472	277	177	26	16
	PrecMin	400	0	0	0	0	2	25	69	123	26	0	0	0
	PrecStd	124	4	1	6	11	23	41	62	76	60	34	5	2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	40.7	42.1	43.0	42.7	41.1	37.9	34.9	36.2	38.8	38.6	37.2
		MCWB	17.6	19.3	20.1	22.0	23.5	25.0	25.9	26.5	26.9	22.3	20.0	19.2
	2%	DB	34.5	39.1	41.0	42.0	41.9	39.9	36.4	33.9	34.9	38.0	37.7	36.0
		MCWB	16.8	19.3	20.3	22.0	23.7	25.2	26.1	26.5	27.0	23.0	20.0	18.4
	5%	DB	33.2	37.7	40.1	41.2	41.1	38.8	35.0	33.0	34.0	37.1	36.9	34.8
		MCWB	16.2	18.7	20.0	21.9	23.7	25.4	26.1	26.3	26.7	23.8	20.0	17.9
	10%	DB	31.9	36.1	39.1	40.1	40.1	37.5	33.7	32.0	33.1	36.1	35.9	33.4
		MCWB	15.7	18.0	19.6	21.7	24.1	25.3	26.0	26.1	26.4	24.2	19.9	17.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	22.6	25.0	26.7	27.2	27.6	27.7	27.7	28.2	28.2	25.0	21.3
		MCDB	31.1	35.1	39.1	38.0	37.2	36.0	34.5	32.7	34.2	35.9	33.9	33.3
	2%	WB	18.1	21.0	23.6	25.6	26.6	27.0	27.1	27.1	27.6	27.3	24.0	19.9
		MCDB	31.8	33.8	35.9	36.7	36.7	35.5	33.4	31.9	33.4	34.6	31.8	32.4
	5%	WB	17.1	19.5	22.8	25.0	26.2	26.6	26.7	26.7	27.2	26.7	23.0	19.0
		MCDB	30.6	35.2	34.3	36.4	36.2	34.9	32.6	31.3	32.6	33.3	30.7	31.5
	10%	WB	16.3	18.5	21.8	24.4	25.7	26.2	26.3	26.3	26.8	26.1	22.0	18.2
		MCDB	30.1	34.8	33.9	35.8	35.8	34.3	32.0	30.8	31.8	32.4	30.6	30.4
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	14.8	14.3	12.3	11.3	10.2	8.6	7.9	8.8	11.5	14.9	15.2
		MCDBR	16.6	16.2	15.2	13.5	12.4	11.2	10.1	9.0	10.0	12.9	15.8	16.7
	5% WB	MCWBR	5.0	5.2	5.4	4.9	3.9	2.9	3.1	3.1	3.6	3.8	3.8	4.3
		MCWBR	15.8	15.0	12.3	11.3	11.0	10.3	9.2	8.4	9.5	10.9	12.8	16.0
(40) Clear-Sky Solar Irradiance	taub	taub	0.524	0.567	0.698	0.716	0.707	0.683	0.608	0.566	0.559	0.584	0.532	0.505
		taud	1.863	1.752	1.519	1.564	1.582	1.661	1.839	1.940	1.943	1.855	1.866	1.909
	Ebn at Noon	Ebn at Noon	773	759	675	662	655	663	718	758	767	736	759	776
		Edh at Noon	199	231	298	283	273	249	210	192	192	205	197	186
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.51	6.05	6.41	6.46	6.28	5.91	5.72	5.55	5.91	5.96	5.66	5.38
	RadStd	RadStd	0.31	0.32	0.33	0.34	0.23	0.17	0.21	0.29	0.29	0.15	0.16	0.21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page